Gb-DS

Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

[18BS3CS03]

# THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2021

## STATISTICS AND PROBABILITY

**Time: 3 Hours** **Maximum Marks: 70**

**Instructions:**
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

**1. a.** Explain different methods of graphical representation of statistical data .Illustrate with example. **07**

**b.** Construct the sub-divided percentage rectangular diagram for the following data. **07**

| Items of Expenditure | Family A (Income Rs. 5000) | Family B (income Rs. 8000) |
|---|---|---|
| Food | 2000 | 2500 |
| Clothing | 1000 | 2000 |
| House Rent | 800 | 1000 |
| Fuel and lighting | 400 | 500 |
| Miscellaneous | 800 | 2000 |
| Total | 5000 | 8000 |

**2. a.** Prices of a particular commodity in five years in two cities are given below **07**

| Price in city A | Price in city B |
|---|---|
| 20 | 10 |
| 22 | 20 |
| 19 | 18 |
| 23 | 12 |
| 16 | 15 |

Which city has more stable prices?

**b.** Calculate Karl – Pearson's coefficient of correlation for the data 25, 15, 23, 40, 27, 25, 23, 25, 20. **07**

**3. a.** From 6 positive and 8 negative numbers, 4 numbers are chosen at random (without replacement) and multiplied. What is the probability that the product is a positive number? **06**

**b.** A Market research firm is interested in surveying certain attitudes in a small **08**

community. There are 125 households broken down according to income, ownership of a telephone, and ownership of a TV.

| | Households with an annual income of Rs. 8,000 or less | | Households with annual income above Rs. 8,000 | |
|---|---|---|---|---|
| | Telephone Subscriber | No Telephone | Telephone Subscriber | No Telephone |
| Own TV set | 27 | 20 | 18 | 10 |
| No TV set | 18 | 10 | 12 | 10 |

i. Evaluate the probability of obtaining a TV owner in drawing at random?

ii. If a household has an income over Rs.8000 and is a telephone subscriber, calculate the probability that he has a TV?

iii. Find the conditional Probability of drawing a household that owns a TV, given that the household is a telephone subscriber?

iv. Are the events 'ownership of a TV' and telephone subscriber statistically independent? comment

**OR**

4. a. If $A$ and $B$ are events with $P(A)=3/4$, $P(B)=1/5$, $P(A\cap B)=1/20$ find the following . $P(A\cup B)$, $P(\overline{A}\cap B)$, $P(A\cap\overline{B})$, $P(A/B)$, $P(B/A)$, $P(\overline{A}/B)$ and $P(\overline{B}/\overline{A})$ **07**

b. In a bolt factory there are four machines $A$, $B$, $C$, $D$ manufacturing respectively $20\%$, $15\%$, $25\%$, $40\%$ of the total production. Out of these $5\%$, $4\%$, $3\%$, $2\%$ are defective. If a bolt drawn at random from the production and is found to be defective. Find the probability that it was manufactured by A or D? **07**

5. a. The probability mass function of a variable $x$ is **06**

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| $p(x)$ | $k$ | $3k$ | $5k$ | $7k$ | $9k$ | $11k$ | $13k$ |

Calculate $k$ ? Also obtain $p(x\le 2)$, $p(x\ge 5)$, and $p(3\le x\le 6)$.

b. For the following probability distribution. **08**

| X | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| P(X) | 1/8 | 3/8 | 3/8 | 1/8 |

Find $E[X]$, $E[X^2]$, $Var[X]$, $SD[X]$, $E[2X]$, $E[2X+3]$, $Var[2X+3]$, $S.D[2X+3]$.

**OR**

6. a. The Kilometre run (in thousands of k.ms ) without any sort of problem in **04**

respect of certain vehicle is a random variable with the following probability density function

$$f(x)=\begin{cases}\frac{1}{40}e^{-x/40} & x \geq 0\\ 0 & x \leq 0\end{cases}$$

Find the probability that the vehicle is trouble free

(a) At least 25,000 k.ms

(b) Between 16000 to 32000 k.ms

**b.** A coin is tossed thrice. Let X be equal to 0 or 1 according as head or tail occurs on the first toss. Let Y be equal to total number of heads which occurs find: **10**

**i.** Marginal distribution of X and Y

**ii.** Joint distribution of X and Y

**iii.** Expectation of X , Y and XY

**iv.** S. Ds of X and Y. (v)Cov (X,Y) and co-efficient of correlation .

**7. a.** Using Newton's forward interpolation formula find the value of $f(1.6)$ **07**

| x: | 1 | 1.4 | 1.8 | 2.2 |
|---|---|---|---|---|
| $f(x)$: | 3.49 | 4.82 | 5.96 | 6.5 |

**b.** Use Lagrange's interpolation formula to find the polynomal $f(x)$ given **07**

| $x$ | 0 | 2 | 3 | 6 |
|---|---|---|---|---|
| $f(x)$ | 648 | 704 | 729 | 792 |

**OR**

**8. a.** Using Newton's divided difference formula, evaluate $f(8)$ . given **07**

| $x$: | 4 | 5 | 7 | 10 | 11 | 13 |
|---|---|---|---|---|---|---|
| $f(x)$: | 48 | 100 | 294 | 900 | 1210 | 2028 |

**b.** Evaluate: $\int_0^6 3x^2 dx$ dividing the interval [0,6] into six equal parts by applying **07**

**i.** Simpson's $(1/3)^{rd}$ Rule

**ii.** Simpson's $(3/8)^{th}$ Rule